Offline Training and Online Inference
A Tiny Neural Stride Prefetcher on 602.gcc_s-734B

602 Stride dynamic inference experiment

Measured ChampSim execution; frozen deployment scope

1 Question and measured answer

How does an offline-trained tiny neural prefetcher behave during a causal, live cache simulation when its weights remain fixed, but its recurrent history and predictions evolve? What inference storage, queueing and service costs accompany its usefulness?

This report reuses 13 completed compatible runs: 6 cold functional runs, 3 frozen h8 finite-resource runs, and 4 historical frozen compatibility runs. Original run identifiers and raw source paths are retained in the committed result tables. These are existing frozen measurements, not relabeled weight-changing runs and not new reruns of the full matrix.

With zero modeled service cost and unbounded PC routing, h8-i1m reaches IPC 0.406066 versus Stride's 0.364976. It saves 6,931,325 cycles over the complete 25M-instruction cold slice, with useful-prefetch coverage 58.05%, selected accuracy 24.37%, legacy timeliness 99.98%, and 1.201 requests per L2 LOAD.

The preserved frozen 16-MAC/cycle case has IPC 0.362900 and saves -391,755 cycles versus Stride (negative means a loss). Only 37.61% of eligible callbacks complete inference.

The preserved frozen 4-MAC/cycle case has IPC 0.362539 and saves -460,332 cycles versus Stride (negative means a loss). Only 20.80% of eligible callbacks complete inference.

Scope. The existing seed-7 i1m/i20m checkpoint is loaded once. Runtime weights remain fixed while h/c and predictions evolve. Offline training memory is excluded from inference deployment.

2 Progression and evidence

Summer work implemented compact PC-keyed LSTM imitation of conventional Stride, with an offline encoder, learned hurdle, positive-count head and signed-delta decoder. The frozen-live experiment moved the same forward model into ChampSim, preserving causal live callbacks. This branch reuses that implementation and preserves the summer code, offline/Colab workflow, checkpoints and frozen raw results.

The regression checks the authoritative Python/C++ frozen forward, changes in h/c and decoded predictions over observed accesses, unchanged parameter values, absence of a training process or update path, finite table eviction, delayed completion without a new demand callback, and the existing cache/metric fixtures. Software checks validate the refactor; the measured tables below retain their original frozen-run provenance.

A fresh paired Sacramento regression executed the 100,000-record development slice after skipping 20M records, using h8-i1m, 64 entries and the 4-MAC service case. The refactored binary and preserved frozen binary matched all 30 original parser fields: 325,790 cycles, IPC 0.306946, 1,633 L2 LOADs, 410 requested prefetches, 2 useful and 0 late. Six snapshot/cohort rows preserved cache and service counters, apart from explicit zero-field initialization and the documented event/queue layout reduction. Host elapsed time was 2.96 s versus 3.07 s; it is not a simulated timing gain. This is software regression evidence, not an additional final scientific row.

3 Causal inference and matched conditions

Only the 64-bit PC and 64-bit aligned byte address enter the neural encoder. A 128-to-H projection and LSTM update h/c for the exact PC. The learned hurdle selects emission; the positive-count head determines K; a GRU decoder emits direct signed cache-line deltas with free-running predicted feedback. Conventional Stride is a separate baseline, retaining its current-line first requests [current, current + stride], tracker behavior and page restrictions. It is not a runtime neural teacher.

The serial inference engine consumes a bounded 16-event input FIFO. A decision uses the last completed h/c for its exact PC. State and outputs become visible at scheduled completion, so queued accesses cannot see a future recurrent state. A finite table has 64 exact-PC entries with LRU replacement; an unbounded control isolates capacity effects. Frozen weights need one active FP32 copy. There is no parameter publication operation.

Historical compatibility preserves the original 25M no-prefetch cache warmup plus 25M measurement, zero modeled neural cost, unbounded PC routing and empty predictor state at the measurement boundary. The actual warmup retirement group ends after 25,000,004 instructions, followed by 25,000,003 measured instructions. The old conventional live Stride run issued throughout warmup and is not used as its matched cycle-saving reference.

The primary protocol is *cold start at the held-out trace-slice boundary.* It skips exactly 25,000,000 opaque records using the actual compiled reader types, without simulating them or warming caches. It executes exactly the next 25,000,000 selected instructions with fresh caches and recurrent histories, loading only pretrained weights. This is not a program-startup experiment. All methods use the same selected records and cache configuration; actual callback order/timing may differ in closed-loop live execution.

Retired target instructions from that boundary are the horizontal axis. Eligible L2 callbacks, admitted decisions and completed decisions are distinct from trace read-ahead and retirement. Snapshots at 0, 1k, 10k and every 100k retired instructions retain startup costs. Curves use exact one-million-instruction counter intervals; cumulative cycle saving always compares matched cold references.

3.1 Assumed inference service

The functional control has zero modeled cost. Sensitivity cases use 4 or 16 MACs/cycle with matched 16 or 64-byte/cycle weight and scratch bandwidth, one nonlinear unit taking four cycles per operation, and serialized dense/nonlinear/memory/control phases. Initiation interval equals data-dependent service time. For hidden size H and actual decoded count K, the forward work is

$$W_{MAC} = 128H + 8H^2 + 2H + eH + K(3H^2 + 4H), \quad W_{NL} = 6H + e + K(3H + 1),$$

where e is the learned emission decision. Control costs 8 + ceil(entries/4) + 4K cycles; additional scratch traffic is 4(128+4H)+24K bytes. The cold decoder resource permits 32 outputs, with excess and numerical failures counted. A 32-address output FIFO releases at most one request/cycle in finite cases through the original request path and duplicate/drop behavior. An unconditional per-core-cycle hook services pending work even when no demand callback occurs. Host execution time is never converted to CPU stall cycles. These are architectural sensitivity assumptions, not measured silicon timing, area, power or synthesized frequency.

4 Cold end-to-end measurements

All rows execute 25,000,000 instructions and observe 404,271 L2 LOAD callbacks. H8/H16 i1m/i20m use the existing seed-7 pretrained weights; they do not represent independent pretraining seeds. State0 denotes the unbounded algorithmic control; state64 is finite LRU. Zero denotes zero modeled neural service cost.

Method	IPC	Cycles	Saved vs Stride	L2 MPKI
h16 frozen i1m	0.406523	61,497,175	7,000,550	3.401
h8 frozen i1m	0.406066	61,566,400	6,931,325	3.419
h8 frozen i1m /64 /MAC0	0.406064	61,566,690	6,931,035	3.419
h8 frozen i1m /64 /MAC16	0.362900	68,889,480	-391,755	8.098
h8 frozen i1m /64 /MAC4	0.362539	68,958,057	-460,332	8.130
h16 frozen i20m	0.406818	61,452,480	7,045,245	3.391
h8 frozen i20m	0.406250	61,538,423	6,959,302	3.444
No prefetch	0.362511	68,963,505	-465,780	8.151
Stride	0.364976	68,497,725	0	6.755

Method	Coverage	Miss reduction	Raw acc.	Selected acc.	Timeliness	P/NL
h16 frozen i1m	0.5828	0.5827	0.2436	0.2436	0.9992	1.2057
h8 frozen i1m	0.5805	0.5805	0.2437	0.2437	0.9998	1.2008
h8 frozen i1m /64 /MAC0	0.5805	0.5805	0.2437	0.2437	0.9998	1.2008
h8 frozen i1m /64 /MAC16	0.0064	0.0064	0.0079	0.0079	0.9985	0.4118
h8 frozen i1m /64 /MAC4	0.0026	0.0026	0.0057	0.0057	0.9887	0.2272
h16 frozen i20m	0.5839	0.5839	0.2751	0.2751	0.9991	1.0698
h8 frozen i20m	0.5775	0.5774	0.2573	0.2573	0.9992	1.1314
No prefetch	0.0000	0.0000	NA	NA	NA	0.0000
Stride	0.1713	0.1713	0.2884	0.2884	0.9952	0.2994

4.1 Raw counter meanings

Let M0 be same-protocol no-prefetch L2 LOAD misses, M method misses, NL L2 LOADs, P requested prefetches, Ipf issued prefetches, Q PQ-merged requests, U useful and L late. The original parser and formulas are preserved:

$$\text{coverage} = U/M0, \quad \text{miss reduction} = (M0 - M)/M0, \quad \text{raw accuracy} = U/Ipf,$$
$$\text{selected accuracy} = U/(Ipf - Q), \quad \text{legacy timeliness} = U/(U + L), \quad \text{pressure} = P/NL.$$

Undefined ratios are NA. Legacy useful may include a first RFO; NL and M retain the original LOAD counter scope. Coverage is not imitation accuracy. Unused eviction and one minus accuracy are not direct cache pollution measurements.

Method	M	P	Ipf	Q	U	L
h16 frozen i1m	85,022	487,444	487,444	0	118,744	99
h8 frozen i1m	85,475	485,458	485,458	0	118,291	25
h8 frozen i1m /64 /MAC0	85,477	485,437	485,437	0	118,288	25
h8 frozen i1m /64 /MAC16	202,455	166,466	166,466	0	1,310	2
h8 frozen i1m /64 /MAC4	203,238	91,858	91,858	0	526	6
h16 frozen i20m	84,784	432,486	432,486	0	118,982	108
h8 frozen i20m	86,101	457,373	457,373	0	117,666	98
No prefetch	203,764	0	0	0	0	0
Stride	168,865	121,020	121,020	0	34,900	168

5 Recorded runtime observations

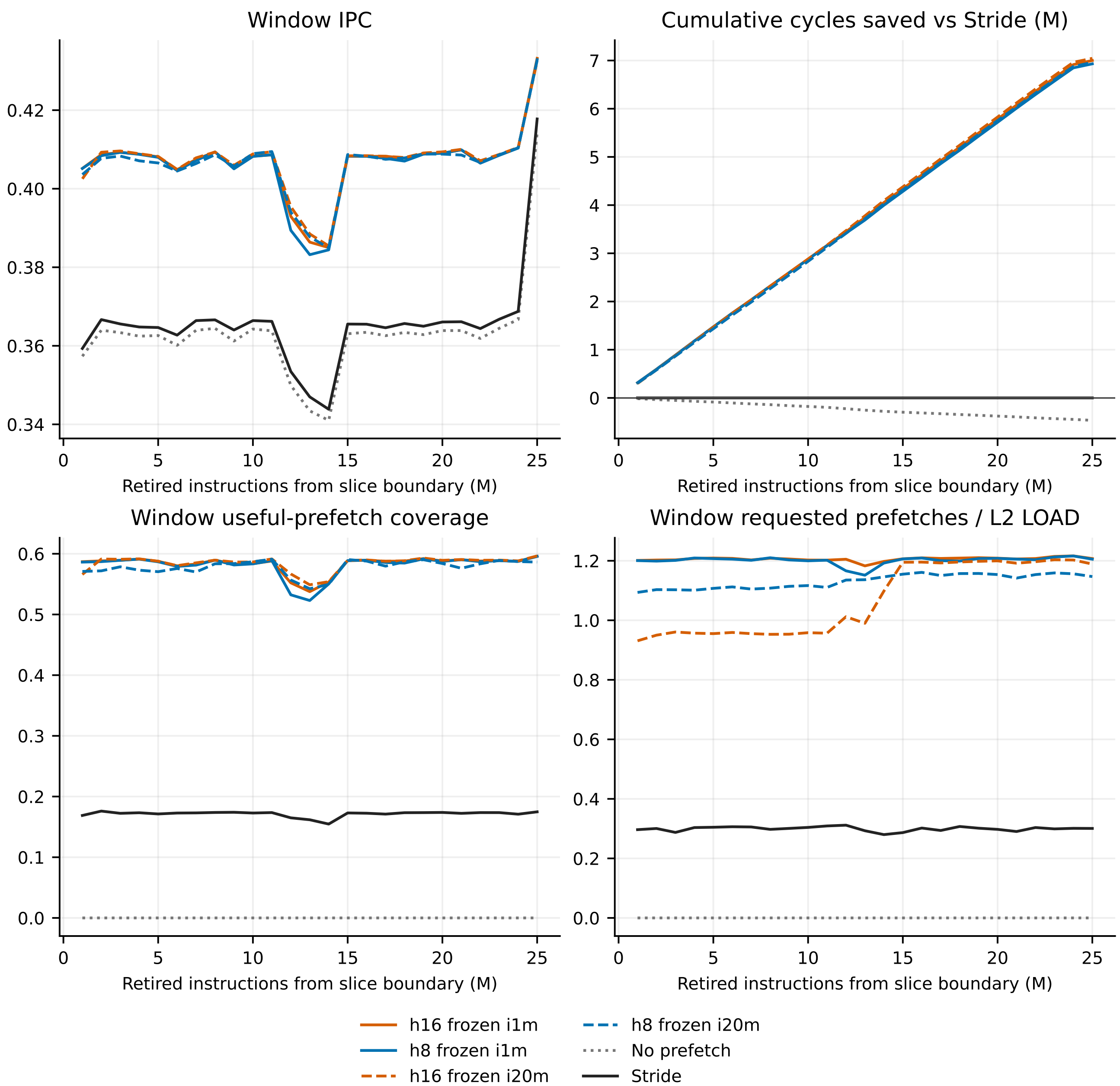
The following h8-i1m zero-cost rows use already recorded cumulative snapshots at 100k, 1M and 25M retired instructions after the 25M trace-slice boundary. Coverage uses NoPF misses at the same recorded instruction count. They are descriptive sample points, not minimum-history requirements, optimized observation budgets or runtime triggers.

Retired instructions	Eligible L2 callbacks	NN completed	Cumulative IPC	Valid L2 lines
100,000	1,659	1,659	0.362541	1,419 / 4,096
1,000,000	16,196	16,196	0.405151	4,096 / 4,096
25,000,000	404,271	404,271	0.406066	4,096 / 4,096

Retired instructions	Coverage	Raw accuracy	Selected accuracy	Timeliness	P/NL
100,000	0.5535	0.3996	0.3996	1.0000	1.1706
1,000,000	0.5863	0.2592	0.2592	1.0000	1.2006
25,000,000	0.5805	0.2437	0.2437	0.9998	1.2008

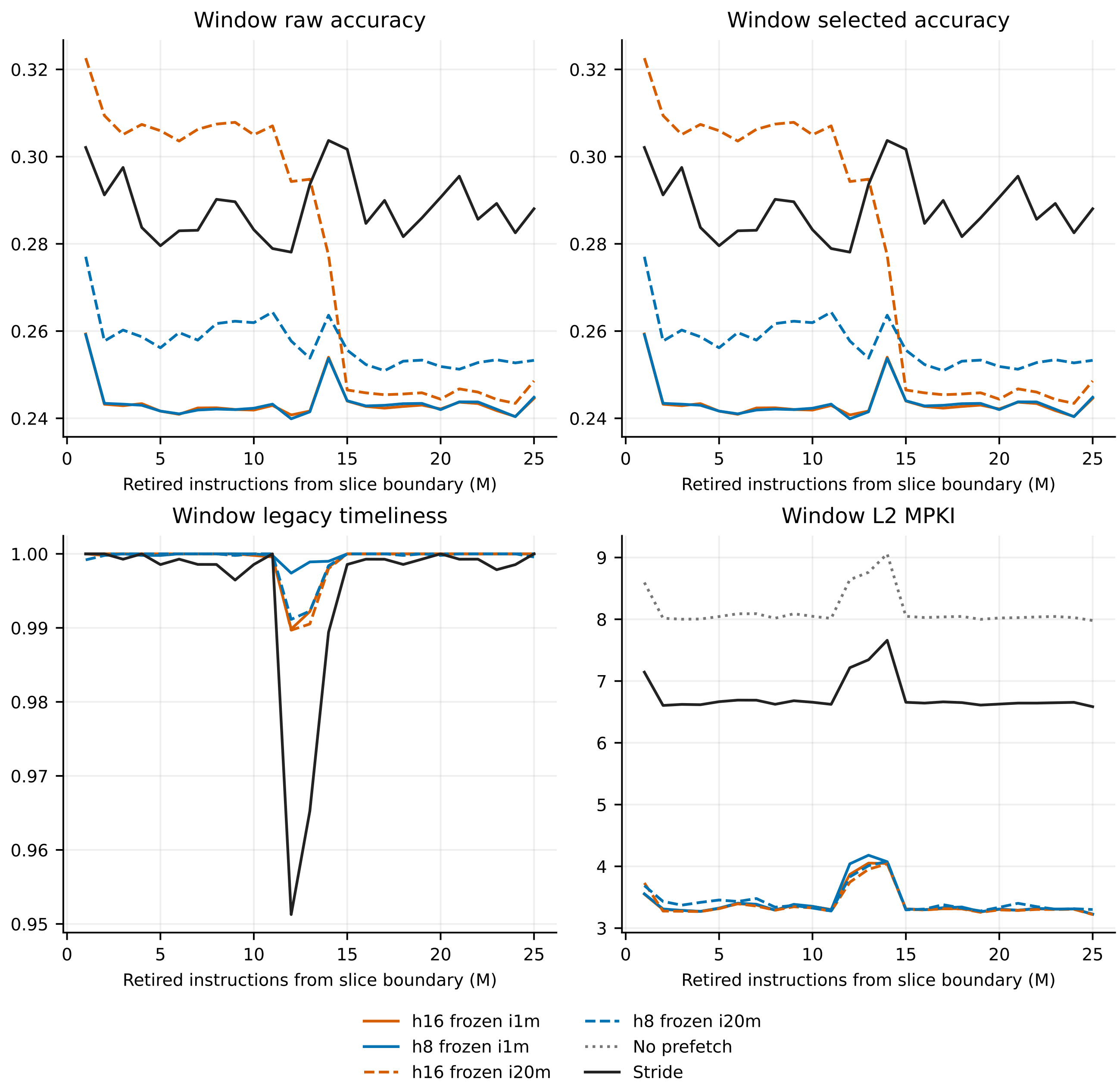
The PC's h/c is carried continuously through successive completed inference events. Snapshot collection and the plotted counter intervals do not reset, truncate or otherwise change that history. No fixed-length history window W is introduced. Weights remain unchanged. The L2 callback count is the actual observation count; at the 1M snapshot it is 16,196, while the legacy completed LOAD counter is 16,195 because callback observation and cache completion have different timing.

6 Dynamic performance with fixed weights



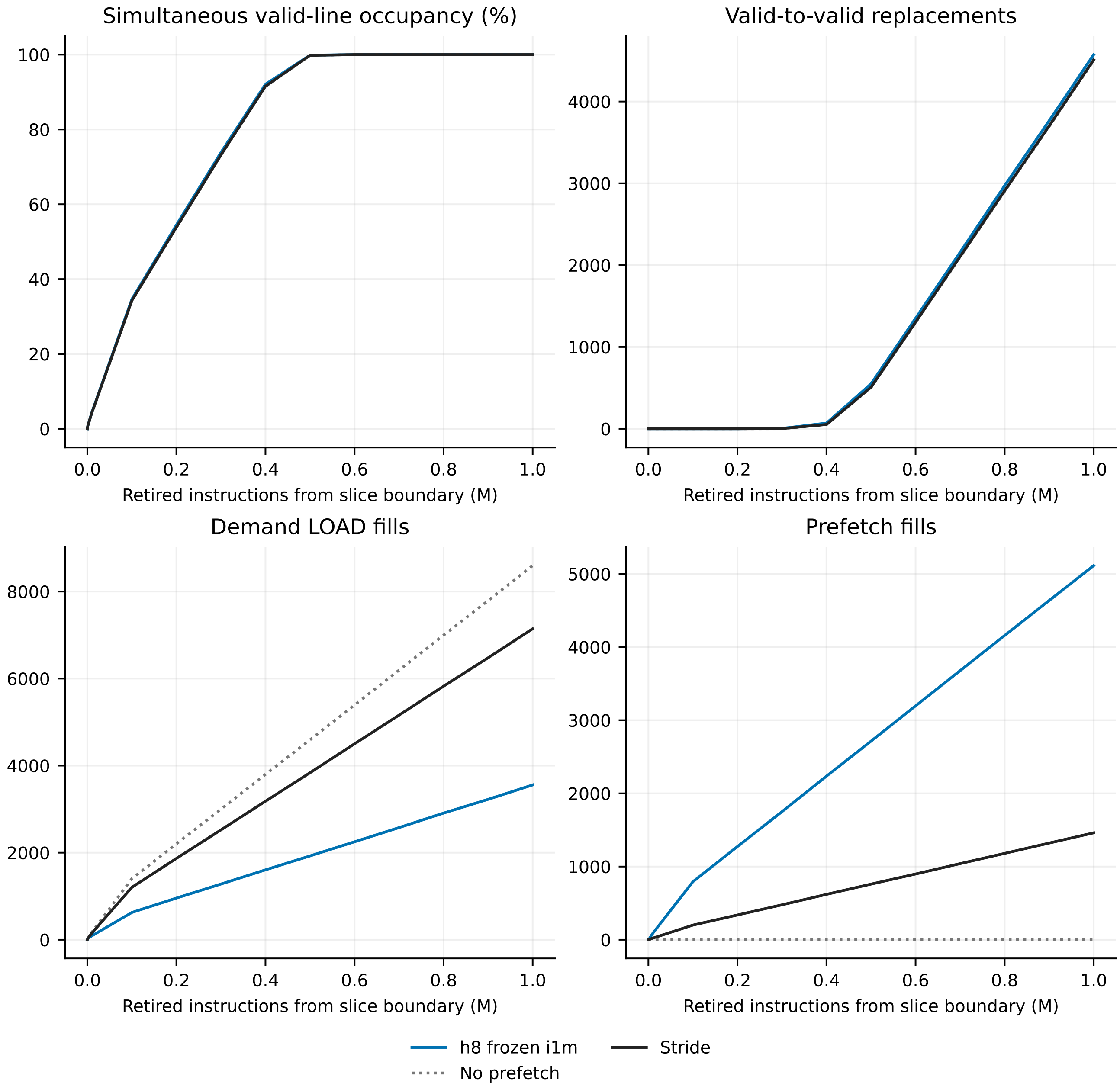
These are live fixed-weight runs. Changing window behavior reflects incoming accesses, evolving h/c, cache state and realized timing; it does not indicate parameter learning. The cumulative curve retains all startup costs. The h8 and h16 i1m/i20m curves are separate checkpoints, with solid i1m and dashed i20m traces.

7 Quality and traffic over execution



A request can cross a window boundary before its first useful demand. These ratios use legacy counter activity within each interval, so a denominator can be zero even when a useful event occurs. NA remains missing, not zero or failure. Separate issue-cohort outcomes retain pending and resident-unused requests as censored. The committed issue-cohort table contains timely, late, unused, redundant and censored counts rather than replacing a legacy metric under the same name.

8 L2 occupancy and turnover



The active L2 has 512 sets, 8 ways and 64-byte lines: 4,096 lines or 256 KiB. Occupancy counts simultaneous valid lines, verified against a scan at samples. Invalid-to-valid fill raises occupancy; valid-to-valid replacement does not. The first million instructions show filling alongside ongoing turnover, not a deadline for prefetch usefulness. Fullness does not establish a warmed working set.

8.1 Exact occupancy attainment and fill outcomes

Method	50% ins.	90% ins.	95% ins.	99% ins.	100% ins.	100% cycles
h16 frozen i1m	177,051	388,585	417,448	460,465	536,006	1,334,339
h8 frozen i1m	176,919	388,585	417,448	460,465	536,006	1,334,484
h8 frozen i1m /64 /MAC0	176,919	388,585	417,448	460,465	536,006	1,334,484
h8 frozen i1m /64 /MAC16	179,239	391,448	419,800	461,325	536,244	1,519,001
h8 frozen i1m /64 /MAC4	179,479	391,448	420,040	461,425	536,244	1,520,031
h16 frozen i20m	175,639	387,882	417,242	460,385	536,006	1,351,583
h8 frozen i20m	177,451	389,020	418,220	460,687	536,006	1,342,677
No prefetch	180,391	392,060	420,440	461,565	536,244	1,520,774
Stride	180,091	391,562	420,040	461,565	536,244	1,512,126

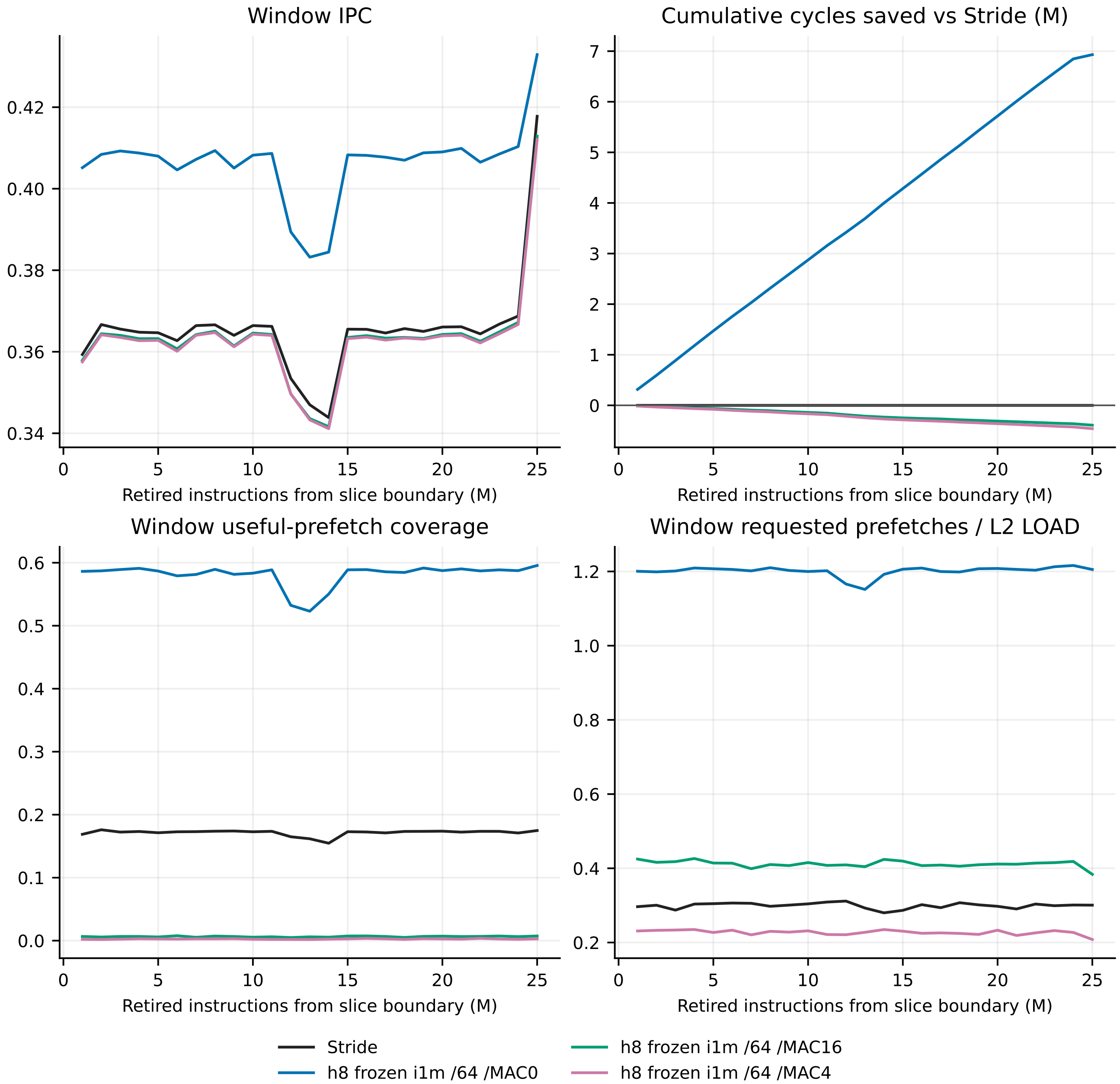
Method	LOAD fills	RFO fills	PF fills	WB fills	Replacements
h16 frozen i1m	85,022	2	122,123	0	203,051
h8 frozen i1m	85,475	2	121,310	0	202,691
h8 frozen i1m /64 /MAC0	85,477	2	121,304	0	202,687
h8 frozen i1m /64 /MAC16	202,455	2	2,933	0	201,294
h8 frozen i1m /64 /MAC4	203,238	2	1,607	0	200,751
h16 frozen i20m	84,784	2	123,592	0	204,282
h8 frozen i20m	86,101	2	122,053	0	204,060
No prefetch	203,764	2	0	0	199,670
Stride	168,865	2	35,439	0	200,210

All listed cold runs reached full occupancy; exact first-attainment instruction/cycle counters are in occupancy_milestones.csv. LOAD, RFO, prefetch and writeback fills retain distinct meanings. Useful line lifetimes and fill-to-first-demand timing are in resources.csv. Request lifecycle observations are measurement-only and are not neural features.

Method	Enqueued	Timely	Late	Unused	Cache red.	Flight red.	Censored
h16 frozen i1m	487,444	118,744	99	3,329	322,363	42,859	50
h8 frozen i1m	485,458	118,291	25	2,974	321,748	42,375	45
h8 frozen i1m /64 /MAC0	485,437	118,288	25	2,971	321,736	42,372	45
h8 frozen i1m /64 /MAC16	166,466	1,310	2	1,607	163,529	2	16
h8 frozen i1m /64 /MAC4	91,858	526	6	1,071	90,241	4	10
h16 frozen i20m	432,486	118,982	108	4,556	264,963	43,823	54
h8 frozen i20m	457,373	117,666	98	4,334	290,092	45,130	53
No prefetch	0	0	0	0	0	0	0
Stride	121,020	34,900	168	535	50,484	34,929	4

Cohorts use unique enqueue and first-demand lifecycle accounting. They are supplementary and need not equal the legacy useful/late counters. Pending/in-flight and resident-unused outcomes remain censored at the final boundary rather than being assigned failure.

9 Finite-service negative results



The frozen 4/16-MAC cases remain below matched Stride end to end. The same finite table at zero cost isolates the table capacity choice. The timed MAC4/MAC16 cases have zero state evictions, so their losses cannot be attributed to h/c eviction. The finite zero-cost control records 30 evictions.

9.1 Inference work, admission and completion

Case	Eligible	Admitted	Completed	Input drops	States peak	Evictions
h8 frozen i1m /64 /MAC0	404,271	404,271	404,271	0	64	30
h8 frozen i1m /64 /MAC16	404,271	152,079	152,062	252,192	63	0
h8 frozen i1m /64 /MAC4	404,271	84,108	84,091	320,163	57	0

Case	MAC/decision	Nonlinear/decision	Service cycles	Input peak	Output peak
h8 frozen i1m /64 /MAC0	1,825.78	78.62	0	1	2
h8 frozen i1m /64 /MAC16	1,801.60	75.92	68,888,246	16	2
h8 frozen i1m /64 /MAC4	1,801.06	75.86	68,958,580	16	2

Work averages use actual started decisions and decoded K. Finite service combines inference throughput, queued state dependencies and address arrival delay. Pending decisions at the endpoint remain pending. Queue pressure and redundant issued requests explain why a useful zero-cost algorithm can lose under these modeled resources. This does not demonstrate that every hardware implementation would have the same service cost.

10 Frozen deployment storage

h8, 64 entries: component	Configured KiB	Observed component peak KiB
active FP32 weights	7.453	7.453
PC tags, metadata and h/c	6.000	5.344
inference scratch	1.312	1.312
active event metadata	0.055	0.055
decoded delta/address payload	0.500	0.031
input queue	0.875	0.875
output queue	0.750	0.047

The h8 model has 1,908 FP32 parameters: 7,632 bytes (7.453125 KiB). H16 has 5,220 parameters: 20,880 bytes (20.390625 KiB). One state entry reserves 32 bytes for exact-PC tag/metadata and 8H bytes for FP32 h/c. Scratch, active event, decoded output and input/output queues are additional deployment storage. There is no neural shadow teacher, training buffer, gradient, optimizer state, training weight copy or publication buffer in frozen deployment.

Method	Configured total KiB	Sum of observed component peaks KiB
h16 frozen i1m	NA	37.078
h8 frozen i1m	NA	17.578
h8 frozen i1m /64 /MAC0	16.945	14.953
h8 frozen i1m /64 /MAC16	16.945	15.680
h8 frozen i1m /64 /MAC4	16.945	15.117
h16 frozen i20m	NA	37.078
h8 frozen i20m	NA	17.578
No prefetch	0.000	0.000
Stride	2.016	2.016

Refactored layout versus retained measurements. The retained scientific runs used a 56-byte active event and a 16-by-56-byte input queue. The frozen-only refactor removes unused event fields, reducing these to 24 bytes and 16-by-24 bytes. Its configured h8/64-entry deployment is therefore 16,808 bytes (16.414 KiB), compared with the original measured layout's 17,352-byte reservation (16.945 KiB): a 544-byte structural reduction. These revised configured bytes are derived from the current packed layout, not retroactively measured allocation peaks. The component-peak and scientific counters above remain those recorded by the original frozen runs.

Unbounded state has no finite configured maximum (NA). Summed component peaks are an envelope, not a simultaneous allocation measurement. Fixed scratch reservations are not allocator measurements. The Stride row includes only its conventional baseline state. Measurement logging and host allocator/RSS overhead are excluded. Dedicated predictor SRAM is compared with the 256 KiB L2 by size; it is not assumed to occupy or pollute L2. These FP32 counts are not quantized deployment measurements.

11 Offline budget and historical frozen compatibility

Offline i1m used 16,172 decisions, 2,486 positive decisions and 4,875 action atoms, with 12 epochs and 228 optimizer steps. I20m used 325,389 decisions, 52,059 positive decisions and 102,140 action atoms, with 12 epochs and 4,404 optimizer steps. These are offline model-preparation observations. They are not runtime observations or fixed-weight deployment allocations.

The previously reported 1M plateau concerns the offline pretraining budget across the completed prefix sweep. It does not mean a frozen runtime learns after one million observations, nor does similar IPC establish identical action behavior. I1m/i20m retain separate rows and their own traffic measurements. The completed historical live logs establish behavior directly; missing logs would not establish equivalence.

Model	Original transport	IPC	Cycles	Coverage	Pressure
h8 i1m	offline keyed replay	0.409124	61,106,147	0.5799	0.8776
h8 i1m	frozen live parity	0.409139	61,103,980	0.5800	0.9519
h8 i20m	offline keyed replay	0.409244	61,088,223	0.5757	1.0162
h8 i20m	frozen live parity	0.409254	61,086,777	0.5757	1.1259
h16 i1m	offline keyed replay	0.409730	61,015,850	0.5832	1.0821
h16 i1m	frozen live parity	0.409738	61,014,571	0.5832	1.2089
h16 i20m	offline keyed replay	0.409938	60,984,797	0.5845	0.9744
h16 i20m	frozen live parity	0.409946	60,983,624	0.5846	1.0726

Preserved compatibility run	Original parser fields exact	Instructions	Cycles
h16 frozen i1m	30/30	25,000,003	61,014,571
h8 frozen i1m	30/30	25,000,003	61,103,980
h16 frozen i20m	30/30	25,000,003	60,983,624
h8 frozen i20m	30/30	25,000,003	61,086,777

Each of the four historical frozen checks matches all 30 original parser fields, 120/120 in total. These are preserved measurements from the predecessor source, with no claim that this refactor newly executed the full historical protocol. The original no-prefetch historical reference has 203,131 measured L2 LOAD misses; it is used only for historical end metrics, never cold windows. Historical warm-cache IPC is not compared against cold-cache IPC as a performance gain.

12 Interpretation and limits

On this one held-out trace slice, offline-trained frozen inference provides useful zero-cost live prefetching with dynamic recurrent state. H8 already captures most of the measured i1m/i20m IPC behavior at lower weight cost than h16. The finite 4/16-MAC results expose queueing and lateness costs that erase that benefit under the stated architectural assumptions. They are retained as negative results. One trace and one pretraining seed do not establish generalization across workloads or statistical equivalence.

The branch's active entry point is run.py; design.md defines the fixed-weight conditions, and command.md records build/test/run/status/resume/report commands and report compilation. Compact raw counters and figure inputs are committed. SPEC traces, checkpoints and full raw logs remain in their existing private trees. All plotted scientific rows are actual frozen or conventional baseline measurements; weight-changing scientific rows are excluded rather than renamed. The raw reference paths continue to identify their original measurements.